

Unforgeable Process Evidence: A New Paradigm of Scientific Integrity Complementing Reproducibility

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Abstract

Reproducibility has become the dominant frame for scientific integrity: a result is trusted insofar as an independent party can rebuild it from shared data and code. We argue that reproducibility verifies *what* was claimed but remains silent about *how* the claim was arrived at, and that this second question is a distinct and complementary axis of integrity. We develop that axis as a set of separable properties—record integrity, temporal commitment, key attestation, capture fidelity, completeness, and process authenticity—and argue that this decomposition constitutes a new paradigm of scientific integrity, complementing rather than replacing reproducibility. We propose *unforgeability* as the mechanizable property at the core of that paradigm: a committed process record that, relative to an explicit threat model and adversary class, cannot be altered, reordered, or backdated without detection. We are careful about what this does *not* establish. Unforgeability secures the record, not the world it describes; it does not by itself show that the underlying events occurred as claimed, nor truth, originality, authorship, or completeness. Its epistemic role is therefore *defeater reduction*: a validated record eliminates a specified class of provenance doubts—post-hoc modification, backdating, silent deletion or reordering, hidden coverage gaps—rather than certifying the conduct itself. We show that the property is attainable today with local, privacy-preserving, cryptographic recording that any third party can verify offline, and we instantiate it in a proof-of-concept exemplar: an open specification (EVIDENCE PACKAGE V1) and a prototype reference implementation (ACADEMIC INTEGRITY RECORDER), released as open source. We close with scope, limitations, and a governance path for adoption.

Keywords: unforgeability; reproducibility; research integrity; process provenance; tamper-evident logging.

1 Introduction

The reproducibility crisis is usually told as a story about *results*: too many published findings cannot be rebuilt from the materials that authors share [Nosek et al., 2015, Donoho et al., 2009]. The standard remedy has therefore been to share more—data, code, protocols, preregistrations—so that an independent party can re-derive the outcome [Nosek et al., 2015, Donoho et al., 2009]. This framing has made *reproducibility* the de facto currency of scientific integrity: a result is trustworthy to the extent that it can be regenerated.

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chong.liu.phil@outlook.com. Code and specification: <https://github.com/chongliuresearch/academic-integrity-recorder>.

We propose a complementary frame. Reproducibility answers the question “*can the result be rebuilt?*” It does not answer “*was the result arrived at honestly?*” A finding can be perfectly reproducible and yet be the product of selective reporting, post-hoc hypothesizing, undisclosed reuse of a collaborator’s work, or an AI-assisted draft presented as original. Conversely, a genuinely honest workflow may fail to reproduce for reasons unrelated to misconduct—fragile materials, unreported environmental conditions, or legitimate stochasticity [Ioannidis, 2005]. The two questions are correlated but distinct, and the crisis literature documents the blind spot explicitly: even when a study is replicated, an observer cannot tell whether the *process* that produced the original was faithful [John et al., 2012, Fanelli, 2009].

What is missing, we argue, is not another call for transparency but a precise, checkable property. We call that property *unforgeability*: relative to an explicit threat model and adversary class, a committed process record is unforgeable if it cannot be altered, reordered, or backdated without that change being detected. Two points bound this claim. First, unforgeability is a property of the *record*, not of the world it describes: it shows that the committed record is intact, not that the underlying events actually occurred. A holder of a valid signing key can always commit a chain of events that never happened; what the keyholder cannot do is rewrite that chain afterward without breaking it. Second, this narrower guarantee is still load-bearing, because the integrity failures that matter for process trustworthiness—post-hoc editing, backdating, silent deletion, hidden coverage gaps—operate precisely by rewriting the record after the fact. Unforgeability thus narrows the ways a claim can be false, without pretending to establish the rest. It gives the broader goal of *process trustworthiness*—verifiable evidence about the conduct of research—a formal, falsifiable, and deliberately bounded core.

The distinction between the goal and the property matters. “Process trustworthiness” describes an aspiration that is easy to affirm and hard to evaluate; “unforgeability” names a property that a skeptic can actually check, because checking it reduces to re-deriving hashes and signatures. A design that aims at unforgeability commits itself to concrete mechanisms—an append-only hash chain, canonical serialization, a device signature, and an external timestamp—each of which can fail or be attacked, and each of which can therefore be stated and scrutinized. This is the sense in which the concept is scientifically productive rather than merely rhetorical.

Our contributions are:

1. a clear separation of *reproducibility* (what was claimed) from *unforgeable process evidence* (how the claim was arrived at), and an argument that the latter constitutes a new paradigm of integrity complementing reproducibility;
2. a decomposition of process evidence into *record integrity*, *temporal commitment*, *key attestation*, *capture fidelity*, *completeness*, and *process authenticity*, together with a threat-model-relative definition of *unforgeability*, its graded character as a *degree of unforgeability*, and an account of its epistemic role as *defeater reduction*—eliminative rather than confirmatory;
3. a positioning of this decomposition against prior provenance, electronic lab-notebook, content-provenance, and proof-of-process systems (§3);
4. a proof-of-concept exemplar—an open EVIDENCE PACKAGE v1 specification and a prototype reference implementation (ACADEMIC INTEGRITY RECORDER), released as open source—demonstrating that the mechanizable properties are achievable with ordinary, local, privacy-preserving tooling;
5. a candid treatment of scope, adversarial limits, and a governance path for adoption.

Non-claim. Unforgeable process evidence proves record integrity—that a committed record has not been altered, reordered, or backdated since it was produced by the attested key, modulo the stated cryptographic assumptions. It does *not* show that the underlying events actually occurred as recorded, nor certify identity, authorship, originality, correctness, or academic integrity, and it cannot prove that all research activity was captured. This boundary is designed into the recording format and its exported manifest, not merely asserted in prose.

2 The Reproducibility Crisis

2.1 What the empirical record shows

The crisis is not anecdotal. In a 2016 *Nature* survey of 1,576 researchers, **70%** reported having failed to reproduce another group’s experiment, and more than half reported failing to reproduce *their own* [Baker, 2016]. Among large coordinated replications, the Open Science Collaboration reran 100 psychology studies and found only **36%** reproduced the original statistically significant result, with effect sizes roughly half as large [Open Science Collaboration, 2015]. In preclinical cancer biology, the Reproducibility Project reproduced only a minority of high-impact findings [Errington et al., 2021], consistent with earlier industry reports that roughly **11%** (6 of 53) of landmark preclinical studies could be reproduced internally [Begley and Ellis, 2012] and that a major pharmaceutical pipeline observed reproducibility in only about one fifth of projects [Prinz et al., 2011].

These numbers are not a verdict on any discipline’s honesty; they are a verdict on the *visibility* of process. As Ioannidis [2005] argued two decades ago, the prevailing incentive structure predicts a large fraction of false positives even without malfeasance, and John et al. [2012] documented that questionable research practices (p-hacking, selective reporting, undisclosed flexibility) are common and often unconscious.

2.2 Why reproducibility alone is insufficient

Reproducibility operates on the *output* side of research. It asks whether, given shared inputs, the same outputs reappear. Three gaps follow.

First, **the reporting gap**. Reproducibility depends on what authors choose to share. Studies show extensive under-disclosure of experimental design, and that the absence of a disclosure norm is itself a cause of non-reproduction [Nosek et al., 2015]. What is never written down cannot be reproduced, and the decision of what to write down is exactly the process reproducibility cannot see.

Second, **the selectivity gap**. Even with full data, reproducibility cannot distinguish a result obtained by pre-specified analysis from one obtained by exploring many analyses and reporting the best [Ioannidis, 2005, Gelman and Loken, 2014]. Preregistration addresses this prospectively [Nosek et al., 2018], but it covers a narrow slice of ordinary research behavior and is silent about the long, messy trail of reading, drafting, and revision that precedes any registration.

Third, **the provenance gap**. Modern research increasingly involves tools—search, writing environments, version control, and AI assistants—whose use is rarely disclosed in a verifiable way [Gao et al., 2023]. Whether and how an AI contributed to a manuscript, whether a passage was copied from a collaborator, or whether a figure was regenerated after peer review are questions about *process*, and reproducibility is structurally blind to them.

None of this diminishes reproducibility; it motivates a second pillar that observes the conduct of research directly rather than inferring it from its outputs.

3 Related Work

The idea that research conduct should be recorded and made verifiable has a long history, and the space between “result” and “process” is more crowded than the crisis framing alone suggests. Three strands are most relevant.

Scientific provenance. The W3C PROV family models provenance as a first-class, machine-checkable object [World Wide Web Consortium, 2013, Moreau and Missier, 2013, Moreau and Groth, 2013], and a substantial literature applies provenance to biomedical data and workflows; one 2024 scoping review alone synthesized 66 studies spanning provenance models, workflow tracking, integrity, reproducibility, and completeness [Gierend et al., 2024]. Electronic laboratory notebooks (ELNs) moved the paper notebook into software [Macrina, 2014], and recent ELN systems such as IntegriLAB combine research-tool integration (GitHub, Overleaf, DataLad) with blockchain-anchored certification of activities and authorship [Easmin et al., 2026].

Content and media provenance. The C2PA standard provides cryptographically signed, tamper-evident manifests for digital media, binding origin, editing history, and AI involvement to content [Coalition for Content Provenance and Authenticity (C2PA), 2024]. It targets media artifacts rather than research conduct, but it establishes the pattern—a signed manifest verifiable without the producer’s cooperation—that this paper applies to the research process itself.

Secure logging and proof-of-process. Tamper-evident append-only logs are well studied [Crosby and Wallach, 2009]. More recent “proof-of-process” work has made explicit the gap that motivates this paper: digital signatures prove key possession, not that the content was produced through a real process, and a signer can always reconstruct and sign intermediate states post hoc [Condrey, 2026].

What the present work adds is not provenance tracking itself—that is well established—but a *decomposition*. Existing systems largely aim to make the record *complete* or *certified*; they rarely distinguish, as we do in §4, the separate epistemic roles of reproducibility, record integrity, capture coverage, and process authenticity, and they rarely make the *non-claims*—that a verified record attests the record, not truth, authorship, or completeness—an explicit, machine-checkable part of the format. Our contribution is that decomposition and its instantiation in a privacy-preserving, gap-aware evidence standard.

4 Unforgeability: Rationale and Epistemic Role

4.1 The warrant of a scientific claim

A scientific result functions as evidence for a claim about the world to the degree that the process producing it can be trusted. That trust decomposes into two questions that are easy to conflate but distinct in what they can be checked. The first asks whether the reported result follows from the reported materials—call this the question of *reproducibility*, of *what* was claimed. The second asks whether the reported process is the *actual* process—whether the experiment, the reading, the analysis, and the writing happened as they are described, in the order described, and before the outcome was known. Call this the question of *fidelity*, of *how* the claim was arrived at. A result can score high on the first question and low on the second, and vice versa.

Science has always treated the second question as load-bearing, even when it lacked the machinery to answer it mechanically. The laboratory notebook is a centuries-old technology whose entire purpose is to make the conduct of experiment visible to later scrutiny, precisely because its contemporaneity—the fact that its entries were written as events unfolded rather than reconstructed afterward—is what gives it evidentiary weight [Macrina, 2014]. The W3C PROV model formalizes provenance as a first-class, machine-checkable object [World Wide Web Consortium, 2013, Moreau and Missier, 2013,

Moreau and Groth, 2013], and open-science movements have long insisted that transparency of process is as important as availability of products [Nosek et al., 2015, Fanelli, 2011]. What these strands share is an implicit commitment to exactly the property we wish to make explicit: that the account of how a result was produced should itself be trustworthy.

The reproducibility crisis can be read as the failure of the present ecosystem to secure this second question. Its signature behaviors—p-hacking, HARKing, selective reporting, the post-hoc construction of a clean narrative—do not primarily attack the regenerability of a result; they attack the fidelity of its provenance, and they survive reproduction precisely because a fabricated-but-consistent artifact can be re-derived just as easily as a genuine one. Reproducibility, by construction, cannot tell the two apart: it verifies the artifact, not the act. If the act is what matters for evidentiary weight, then a property that binds the act—rather than the artifact—is what is missing.

4.2 Unforgeability as the operative property

A record of the research process can be asked to vouch for several different things, and these must be kept apart, because they have different statuses and different mechanisms. We distinguish six properties:

- **record integrity** — the committed record has not been altered, reordered, or truncated without detection;
- **temporal commitment** — the record, or a hash of it, existed no later than some externally attested time;
- **key attestation** — the record was produced by the holder of a specific key or device;
- **capture fidelity** — recorded events correspond to the external events they claim to describe;
- **completeness** — all relevant activity was captured; and
- **process authenticity** — the recorded process is the process that actually occurred.

The first three are mechanizable with standard cryptography; the last three are not, in general, mechanically verifiable at all. Capture fidelity holds only relative to a trusted capture boundary, and completeness and process authenticity are beyond what any recording system can prove about its own inputs. A signer can always commit a chain of events that never happened; no amount of hashing or signing can detect that, because the fabrication happens *before* recording, not after. This gap between key possession and process authenticity is precisely what recent “proof-of-process” work has made explicit [Condrey, 2026].

We reserve *unforgeability* for the first property, stated precisely. Relative to a threat model T and an adversary class $\mathcal{A}$, a committed record is *unforgeable* if $\mathcal{A}$ cannot alter, reorder, or back-date it without that change being detected by re-deriving its hashes and signatures. So defined, unforgeability is a property of the *record*, not of the world it describes: it certifies that the record has not been rewritten since it was committed and signed, and nothing more. It is this precise, checkable property—not the unattainable one of “the process really happened”—that the rest of the paper adopts. Two qualifications keep the claim honest. First, unforgeability is *computational* and *conditional*: it rests on the hardness of standard primitives—collision-resistant hashing, unforgeable signatures under the Ed25519 scheme [Bernstein et al., 2012], and authenticated encryption [Bernstein, 2008]—and on operational assumptions such as the secrecy of signing keys and the integrity of any external timestamp. Second, it is a property of the record, not a verdict on the research: an unforgeable record of a flawed experiment is still a flawed experiment.

This precision is what gives the property its epistemic force, and we state it as *defeater reduction*. A scientific claim faces many possible defeaters—ways in which the evidence for it could fail. A verified record does not eliminate all of them; it eliminates a *specified class*, namely those that operate by rewriting the record after the fact: post-hoc editing, backdating, silent deletion or reordering, and hidden coverage gaps. Those defeaters are exactly the ones reproducibility cannot see, because they attack the act rather than the artifact. Converting them from matters of contested testimony into matters of checkable fact is a modest but genuine strengthening of the evidence base—modest, because it never establishes truth; genuine, because it narrows the ways a claim can be false. The exemplar in §5 is organized around exactly this property: a hash chain, canonical serialization, a device signature, and an optional timestamp, each contributing one of the three mechanizable properties.

4.3 Unforgeability as a matter of degree

We have treated unforgeability as a cryptographic property, but the underlying notion is more general, and recognizing that generality clarifies both what the cryptography buys and why the concept is epistemically fundamental.

Unforgeability, broadly understood, attaches to *evidence* and to the specific claims that evidence is meant to secure, and it is a matter of *degree* rather than an all-or-nothing property. Consider two photographs of the same scene. One is a yellowed, creased print offered by an elderly woman in a remote mountain village; the other is a visually identical image supplied by a young engineer in a high-tech industrial park. As images they are indistinguishable, but as evidence they are not equally credible. The difference lies not in what the images show but in how hard it would be, for each bearer, to produce a convincing counterfeit. The village photograph is embedded in circumstances—physical aging, the absence of any fabrication infrastructure, and a bearer with neither the means nor the motive to forge—that together make a counterfeit expensive and improbable. The engineer’s image can be synthesized or altered with tools and skill that are cheaply available to its bearer. The first photograph has the higher *degree of unforgeability*, and that is what underwrites its greater evidentiary weight.

This graded, pre-theoretic notion is exactly what the cryptography of §5 makes explicit and, within a bounded domain, near-maximal. A hash chain and a device signature add no new *content* to a record; they raise its degree of unforgeability by attaching a checkable difficulty to any after-the-fact counterfeit. The degree is never absolute—it is relative to a threat model and to the resources and assumptions granted to an adversary—but it is made *explicit*, where the village photograph’s degree is merely implicit and circumstantial. This, we suggest, is the deeper point: the formation of evidence, and therefore of trustworthy knowledge, has always depended on this largely unspoken notion of how hard a thing is to forge. Process evidence makes that notion explicit, checkable, and precise within its stated bounds.

A corollary follows. The epistemic contribution of process evidence is primarily *eliminative* rather than directly *confirmatory*. A validated record does not establish that the scientific claim is true, or even that the entire research process occurred exactly as represented. It rules out, or substantially weakens, a specific class of alternative explanations—that committed records were silently altered, reordered, or deleted after the fact. It removes grounds for doubt rather than supplying positive proof of honesty or truth. That is a real and measurable gain, but it is a gain in the strength of a *warrant*, not a shortcut to certainty.

4.4 Complementarity with reproducibility

Unforgeability is not a substitute for reproducibility; the two secure different questions, and both are needed. Two clarifications guard against over-reading the relationship. First, reproducibility is

itself narrower than is often assumed: in the standard usage it means that a reported result can be recomputed from the same data, code, and procedures—it establishes recomputability, not truth, and not the soundness of the process that produced the data. Second, a verified process record reduces a *specified class* of provenance doubts; it does not convert a failure to reproduce into a purely methodological verdict. A result can be reproducible and yet rest on a fabricated or selectively edited account, and a result can be unforgeably recorded and yet be wrong, dishonest, or built on events that were themselves fabricated before recording.

The relationship is cleanest as a two-by-two space (Table 1). A result that is reproducible but whose process record is forgeable can be rebuilt yet may rest on a fabricated account. A result whose record is unforgeable but which fails to reproduce still faces open questions of method, honesty, and unrecorded activity—its record is intact, but its inference remains doubtful. A result that is both reproducible and unforgeably recorded enjoys the strongest available evidentiary basis short of independent corroboration of the underlying phenomenon. And a result that is neither commands the least.

		Reproducible	Not reproducible
Unforgeable process	pro-	result regenerates <i>and</i> the record is intact (strongest)	record intact but result does not regenerate (provenance doubts reduced; method, honesty, or unrecorded activity remain possible)
	Forgeable process	result regenerates but process may be fabricated or edited	neither regenerable nor attested (weakest)

Table 1: Unforgeability and reproducibility secure different questions and are independent. Each cell indicates what the two properties jointly establish.

Two observations follow. First, the properties are *independent*: knowing that a result reproduces tells us nothing about whether its process was genuine, and knowing that its process was unforgeably recorded tells us nothing about whether the result will reproduce. This independence is exactly why unforgeability contributes evidence that reproducibility cannot supply. Second, the value of unforgeability is asymmetric and incremental: it does not raise the ceiling of what science can claim, but it raises the floor below which evidence cannot easily fall. By making the retroactive rewriting of the record detectable, it converts a category of integrity failures from matters of contested testimony into matters of checkable fact. That is a modest but genuine strengthening of the evidentiary basis on which scientific truth-claims rest—modest, because it never establishes truth; genuine, because it narrows the ways a claim could be false.

4.5 Technical feasibility

The property would be idle if it were not attainable with ordinary tooling, on ordinary machines, without imposing surveillance. Three developments make unforgeable process evidence practical now where it was not a decade ago.

Local-first capture without a panopticon. The evidence can be collected on the researcher’s own machine, with no server, no telemetry, and sensitive content never leaving the device in cleartext. This addresses the principal objection to process recording—that it is surveillance—and aligns with data-minimization norms [Voigt and Von dem Bussche, 2017]. Local-first is a design choice in service of unforgeability’s adoption, not a headline property: it removes the single largest reason a researcher would refuse the tool, and thereby widens the conditions under which an unforgeable record can exist at all.

Cheap, standard cryptography. Every primitive the property needs is library-grade. An append-only hash chain commits each event to its predecessor, so deletion, reordering, or editing breaks the chain [Crosby and Wallach, 2009]. Canonical serialization (RFC 8785 JCS) makes hashing independent of key order and whitespace [Rundgren, 2020]. A device signature under ED25519 attests the recorder and gives the record its unforgeable anchor [Bernstein et al., 2012]. Authenticated encryption (XCHACHA20-POLY1305) seals sensitive content [Bernstein, 2008]. An internet-independent timestamp, such as OpenTimestamps, can later prove existence at or before a point in time without trusting a notary [OpenTimestamps, 2018]. None of this is exotic; all of it is battle-tested.

Offline, adversarial verification. Because the property is mathematical, a skeptic needs no special access to check it. The verifier is a small, auditable program that re-derives every hash and signature; its success, rather than any party’s assertion, is what licenses the inference that the record is genuine. This is the same principle that makes signed commits and checksums trustworthy without a trusted third party, and it is what distinguishes unforgeable process evidence from a self-reported methods section, which remains forgeable however detailed it is.

5 A Feasibility Exemplar: a Prototype Reference Implementation

We now illustrate—not exhaust—the argument with a deliberately brief account of ACADEMIC INTEGRITY RECORDER, an open-source *prototype* reference implementation of the EVIDENCE PACKAGE v1, released under the AGPL-3.0 license and open to community contribution. It is a proof-of-concept, not a finished system: the specification is the stable artifact, while the code is an evolving demonstration that the property is attainable, with deliberately partial platform support. We therefore describe only the mechanisms that carry the argument—*unforgeability*, *local-first capture*, *honest capability disclosure*, *privacy-by-default*, and *offline verification*—and leave implementation detail to the project itself.

5.1 Architecture

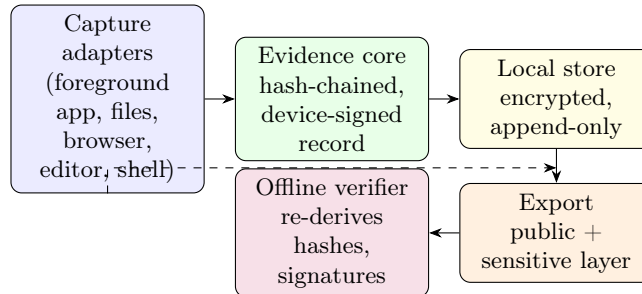


Figure 1: Sketch of the architecture. Evidence flows from capture adapters into an evidence core that maintains a hash-chained, device-signed record, is persisted in an append-only local store, exported as a two-layer EVIDENCE PACKAGE v1, and verified entirely offline. No telemetry leaves the device.

Capture adapters observe activity—foreground application, file events, browser and editor interactions, and (opt-in) shell commands. They forward events to the evidence core, which canonicalizes, hashes, signs, and persists them (Figure 1).

5.2 The hash chain

The core mechanism is an append-only hash chain. Each recorded moment is an *evidence event* carrying a sequence number, timestamps, a source, a semantic kind, and a sensitivity label; its payload hash and the hash of the preceding event are committed into the event’s own hash. Because every link binds its predecessor, editing, deleting, or reordering any event changes its hash and so breaks every subsequent link and the signed head of the chain [Crosby and Wallach, 2009]. The primitives are standard and battle-tested: collision-resistant hashing, a device signature, authenticated encryption for sensitive content, and an optional internet-independent timestamp to prove existence at or before a point in time [Rundgren, 2020, Bernstein et al., 2012, Bernstein, 2008, OpenTimestamps, 2018]. Events are persisted as immutable, append-only encrypted segments with signed checkpoints of the chain head; a crash is reconciled on open, and unsigned, missing, or reordered segments are rejected, so history cannot be silently rewritten.

5.3 Export and honest disclosure

Export produces a single EVIDENCE PACKAGE V1 with a signed manifest and two layers. The *public layer* carries the event chain with payloads replaced by their hashes and a human-readable report, so integrity can be checked without revealing content; the *sensitive layer* holds the full events, artifacts, and disclosures, sealed under a passphrase-derived key. The manifest states exactly what the package claims—and, just as important, what it does not.

The design’s distinctive mechanism is *honest capability disclosure*. Each capture capability is reported as AVAILABLE, PERMISSIONREQUIRED, DEGRADED, or UNAVAILABLE, and any lapse in coverage is recorded as an explicit **Gap** event rather than silently omitted. A research claim can also be *anchored* to a specific passage of a manuscript and later revalidated, and an *active-time* measure reports a conservative lower bound on observed foreground activity. This honesty is what makes the record trustworthy: it is explicit about where it stops.

5.4 Offline verification

Because the property is mathematical, a skeptic can check it without special access: a small, auditable verifier re-derives every hash and signature from the package alone, and its success—not any party’s assertion—licenses the inference that the record is genuine. Success proves *record integrity and a device signature only*, and the verifier’s own output says so.

6 Scope, Limitations, Governance, and Adoption

6.1 Scope and applicability

Unforgeable process evidence is most valuable where the contested question is *how*, not *whether*: disclosure of AI assistance, provenance of a passage, the trail of revision, the existence of a planned experiment that was later dropped, or a contemporaneous lab record. It strengthens theses, grant reports, and post-publication audit alike.

6.2 Limits of the property

Unforgeability is powerful but bounded, and the bounds belong to the property itself, not to any particular implementation.

- **Computational and conditional.** The property holds only while signing keys remain secret and any external timestamp is honest; a compromised key or a controlled clock can retrospectively undermine the record. Unforgeability is a guarantee under assumptions, not an absolute.
- **Attests the record, not the world.** An unforgeable record shows that the committed record has not been altered after commitment. It does not establish that the recorded events correspond to real occurrences, that the result is true, the method correct, or the work original; a faithfully recorded but flawed experiment remains flawed.
- **No completeness.** The property secures what was *recorded*, not that *everything* was recorded. Activity outside the instrumented environment is invisible to it, and no design can convert an absence of evidence into evidence of absence.
- **Not identity or authorship.** A device signature attests a key, not a legal person; it cannot distinguish the key’s owner from anyone else using the device or account.
- **Cannot close the environment.** A determined adversary may work on uninstrumented machines or fabricate inputs before they are recorded; the property narrows the ways a claim can be false but does not eliminate every one.
- **Complements, not substitutes, review.** The record is evidence to be weighed alongside reproducibility and peer review, never a verdict on merit.

These limits are features when stated plainly: unforgeable evidence is trustworthy *because* it is explicit about where it stops.

6.3 Governance and adoption path

We propose unforgeable process evidence as a lightweight, open community standard, in the spirit of how code signing and transparency logs became default expectations [Crosby and Wallach, 2009]. Concrete steps:

1. **Open specification and interop.** The format and its verification are specified openly, so any archive or journal can check a deposited package without vendor lock-in.
2. **Complement, don’t replace.** Tie the record to existing norms: the data-availability statement, the preregistration, and the conflicts/disclosure form. The AI-disclosure entity maps directly onto emerging AI-use policies [Gao et al., 2023, Committee on Publication Ethics (COPE), 2023].
3. **Tiered adoption.** Start with self-archiving and reviewer-on-request, then optional journal encouragement, then (where appropriate) a deposition requirement for certain claim types. Voluntary, low-friction adoption avoids the surveillance objection.
4. **Incentives for honesty.** Because gaps are recorded, a complete, gap-light record becomes a positive signal; a record full of disclosed gaps is at least honest, which is more than the current default of silence.

7 Conclusion

Reproducibility verifies *what* was claimed; unforgeable process evidence verifies *how* the claim was arrived at—but only in a deliberately bounded sense. We have argued that this second axis is a new paradigm of scientific integrity—not a replacement for reproducibility but a partner to it, closing the gap that reproducibility, by construction, cannot see. Its precise content is defeater reduction: a verified record does not prove that the recorded process occurred, or that the science is sound; it makes a specified class of retrospective manipulations—editing, backdating, silent deletion, hidden gaps—independently detectable, while remaining explicit about truth, authorship, and completeness, which it does not establish. The property is attainable today with local, privacy-preserving, cryptographic recording and offline adversarial verification, as an open specification and a prototype implementation demonstrate. Adopting unforgeable process evidence will not end the reproducibility crisis, but it gives the scientific community a new, precisely bounded instrument for the part of integrity that reproducibility was never designed to observe.

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